Swiggy Sales Analysis

End-to-end SQL project — data cleaning, star schema, KPIs & deep-dive analytics

BY ABHAY

SQL SERVER · T-SQL

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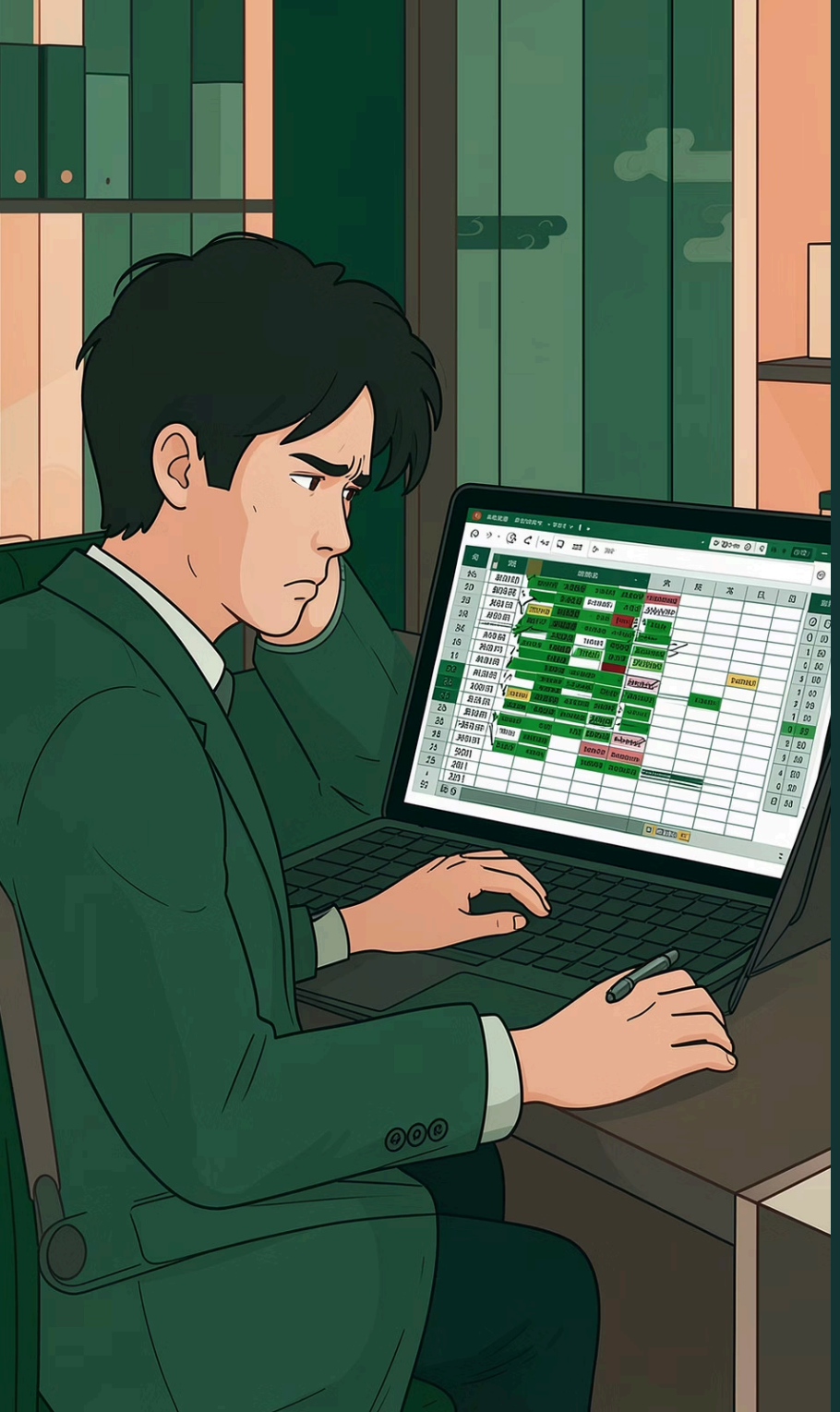

The Problem

Raw delivery records — all in **one flat table** with nulls, blanks & duplicates

Cities & Revenue

Top performers?

Food Trends

Popular categories?

Order Patterns

Monthly & weekly trends?

Pricing & Ratings

Avg spend & satisfaction?

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Four-Phase Solution

Phase 1: Validation

Phase 2: Star Model

Phase 3: KPI Dev

Phase 4: Deep Analysis

Each phase builds on the last — from raw, messy data to structured, query-ready analytics.

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Phase 1 — Data Cleaning

Null Check

Conditional aggregation across 10 business-critical columns

Blank String Filter

Catch empty strings missed by standard NULL checks

Duplicate Removal

CTE + `ROW_NUMBER()` — keep first, delete surplus

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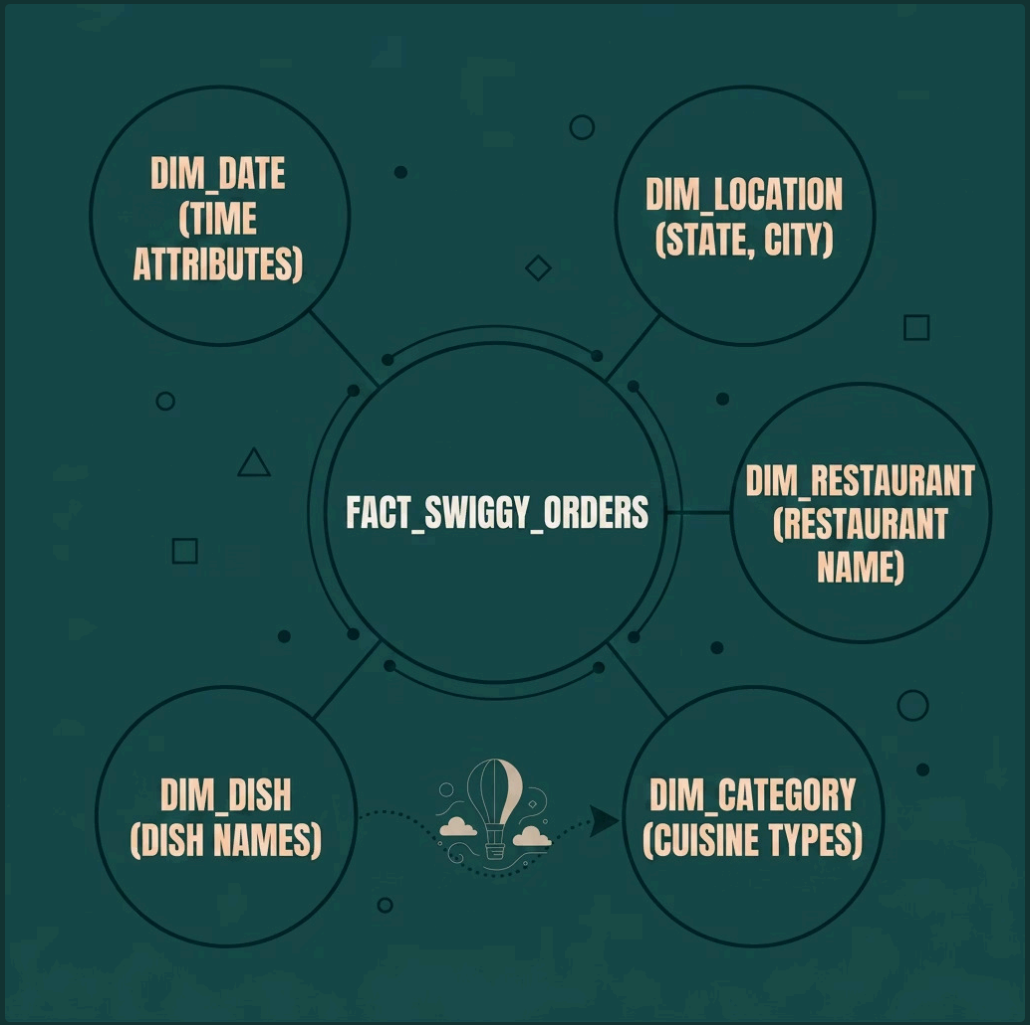
Phase 2 — Star Schema



Why Star Schema?

- Eliminates redundancy — names stored once
- Faster queries — small focused lookups
- BI-ready — Power BI, Tableau, Excel

5 dimension tables + **1 fact table** with enforced FOREIGN KEY constraints

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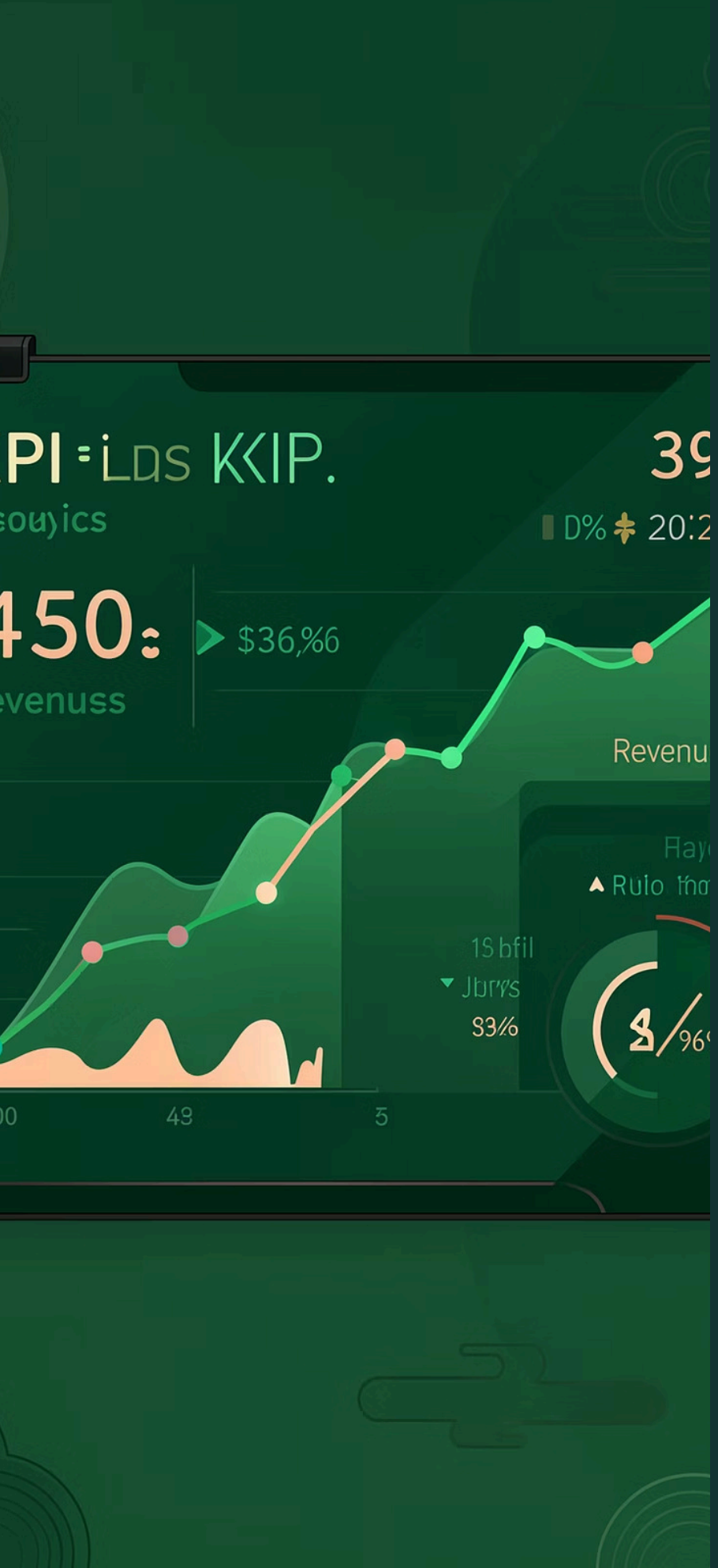

Phase 3 — Core KPIs

Total Orders

`COUNT(*)` on fact table

Total Revenue

`FORMAT(SUM/1000000, 'N2')` INR Million

Avg Dish Price

`AVG(CONVERT(FLOAT))` INR

Avg Rating

Platform-wide satisfaction benchmark

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Phase 4 — Time & Location Analysis

Time Trends

- Monthly orders & revenue
- Quarterly growth
- Year-over-year platform growth
- Day-of-week peak patterns

Location Insights

- Top 10 cities by order volume
- Revenue by state

Queries use `TOP N`, `DATENAME` + `DATEPART` for correct weekday sort

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Food & Restaurant Performance

Top Restaurants

Top 10 by revenue — partnership priority

Top Categories

Most profitable cuisine types by revenue

Most Ordered Dishes

Top 10 dishes — menu recommendations

Cuisine Performance

Orders + Avg Rating — find hidden gems

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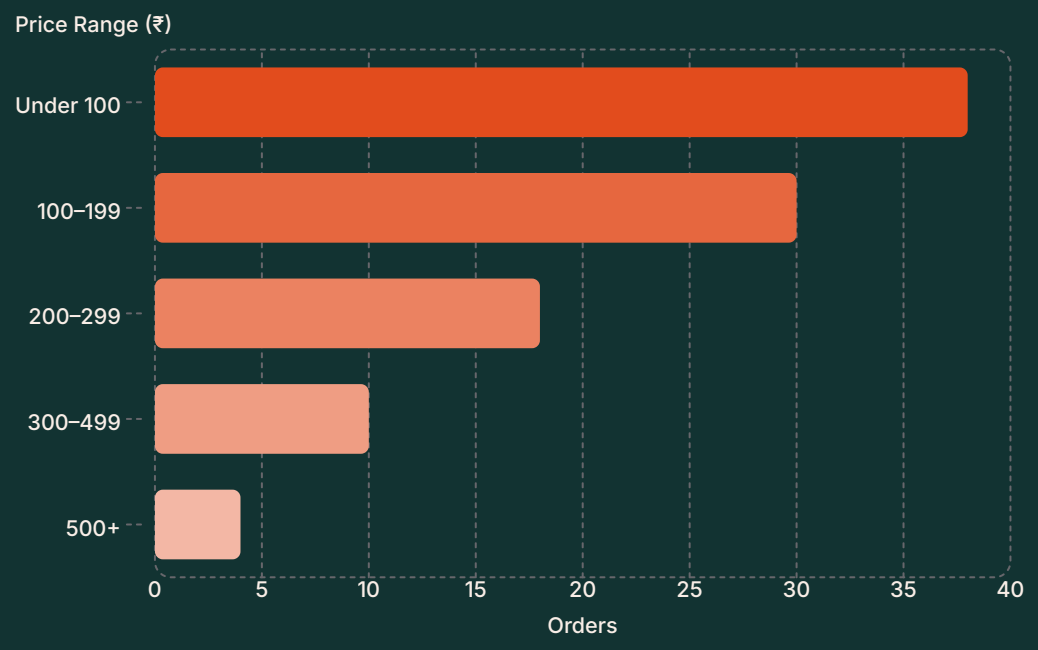
Spending & Ratings



Price Range Distribution

Majority under ₹100 — budget-conscious platform

Rating Distribution (1–5)

Grouped by rating — platform satisfaction spread

CASE WHEN buckets inside subquery for segmentation

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Key Takeaways

01

End-to-End Workflow

Raw data → clean → star schema → KPIs → 12+ queries

02

Advanced T-SQL

CTEs, window functions, multi-table JOINs, subqueries

03

BI-Ready Design

5 dims + 1 fact table — Power BI & Tableau ready

04

Business Impact

Real questions — cities, revenue, trends, pricing, ratings

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